NUMBER SYSTEM MADE EASY | VIDEO 4

Decimal → Binary

Using the Division-Remainder Method



Quick Review

Decimal

Base 10

Binary

Base 2

Today: Decimal → Binary

The Method: 5 Simple Steps

1. Divide the number by 2
2. Record the remainder
3. Divide the quotient by 2 again
4. Continue until the quotient becomes 0
5. **Read the remainders from bottom to top**

Example 1: Convert 45_{10} to Binary

45 ÷ 2 = **22**	R 1
22 ÷ 2 = **11**	R 0
11 ÷ 2 = **5**	R 1
5 ÷ 2 = **2**	R 1
2 ÷ 2 = **1**	R 0
1 ÷ 2 = **0**	R 1



READ BOTTOM → TOP

101101

$45_{10} = 101101_2$

Why Read Bottom to Top?

Division

Top → Bottom



Reading the answer

Bottom → Top



First remainder → smallest positional value

Final remainder → largest positional value

Example 2: Convert 25_{10} to Binary

25 ÷ 2 = **12**	R 1
12 ÷ 2 = **6**	R 0
6 ÷ 2 = **3**	R 0
3 ÷ 2 = **1**	R 1
1 ÷ 2 = **0**	R 1



READ BOTTOM → TOP

11001

$25_{10} = 11001_2$

Your Turn!

Convert:

$$13_{10} \rightarrow ?_{2}$$

You have 20 seconds!



Answer: 13_{10} in Binary

13 ÷ 2 = **6**	**R 1**
6 ÷ 2 = **3**	**R 0**
3 ÷ 2 = **1**	**R 1**
1 ÷ 2 = **0**	**R 1**



READ BOTTOM → TOP

1101

$13_{10} = 1101_2$

Recap: Decimal → Binary

1. Repeatedly divide by 2
2. Record every remainder
3. Stop when the quotient = 0
4. Read the remainders bottom → top

Next video: Binary ↔ Hexadecimal